Lecture 7
Coupled bending-twist for non-rotating blades

ASDA/Dynamics of Rotor Systems

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This lecture

- Coupled and uncoupled PDEs
- Partial Differential Equations of Motion for inertially coupled bending-twist of non-rotating blade
- Other examples of coupled problems

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Previous uncoupled PDE models

Uncoupled problems = bending and twist (torsion) statics
Example of uncoupled bending and twist statics:

$$\left(EI(x)\,w''(x)\right)'' = q(x)$$ *No θ(x) in this equation*

$$\left(GJ(x)\,\theta'(x)\right)' = -t(x)$$ *No w(x) in this equation*

Uncoupled problems = flap and lead-lag for rotating blade
Example of uncoupled flapping bending and lead-lag bending dynamics:

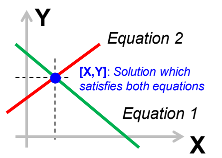
$$\left(EI_F\, w''\right)'' - \left(T\, w'\right)' + m\ddot{w} = L$$ *No v(x,t) in this equation*

$$\left(EI_L\, v''\right)'' - \left(T\, v'\right)' + m(\ddot{v} - \Omega^2 v) = D$$ *No w(x,t) in this equation*

Coupled problems = what is it?
When equations (of motion) can not be solved independently.

$$X + Y = 2$$
$$Y - X = 1$$

Both equations (problems) have to be solved at the same time.



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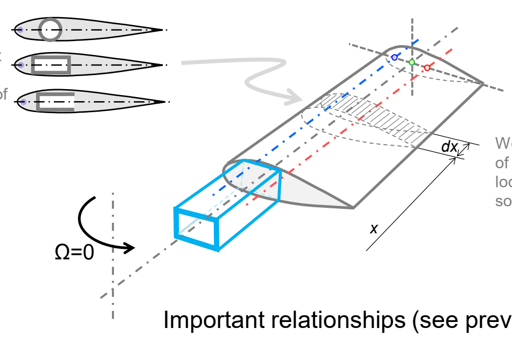
Introduction to bending-twist (non-rotating blade)

To make the derivation simpler, we consider non-rotating blades, Ω=0. The adopted approach is also applicable to *slender* wings in fixed-wing aeroelastic applications.

Sectional constants reflect and represent the bending and torsional properties of the sections.



We study dynamic equilibrium of a general *blade element* (not located on the boundary of the solved domain).

Important relationships (see previous slides):

$$\frac{dM}{dx} + S - \cancel{T\frac{dw}{dx}} = 0$$

These relationships and equations are typically obtained as a byproduct of the PDE derivation for any given problem. They support the BC construction too.

Bending	**Twist**	
$M'_F + S_F = 0$	$Q' - M_\theta = 0$	… *differential* equilibrium
$S'_F = F_{I,F} - L$	$M_\theta = M_I - M_A$	… applied loads
$F_{I,F} = m\ddot{w}$	$M_I = I_{e.c.}\ddot{\theta}$	… local inertia loads
$M_F = EI_F\, w''$	$Q = GJ\,\theta'$	… constitutive models

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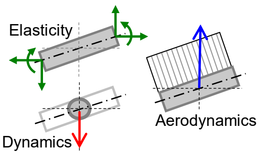
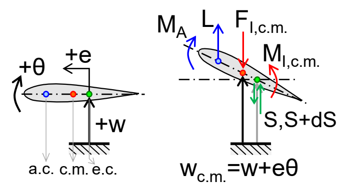
Bending-twist problem – bending equation

Assume a non-rotating blade modelled as beam (1D continuum) where each point in the beam's domain has assigned a bending w(x,t) and twist θ(x,t) deformation. As before, consider the dynamic equilibrium of an arbitrary blade segment *dx* located within the beam domain, i.e., *eR<x<x+dx<R*:

Arbitrary blade segment FBD – front view



Arbitrary blade segment FBD – side view



Differential bending equilibrium from previous slides:

$$M' + S = 0$$

Vertical force equilibrium:

$$-S + S + dS + L\,dx - m\,dx\,(\ddot{w} + e\ddot{\theta}) = 0$$

Combining these two equations yields the bending equation due to bending and twist:

$$M'' + m(\ddot{w} + e\ddot{\theta}) = L$$

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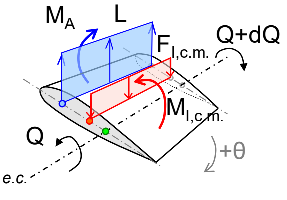
Bending-twist problem – twist equation

Assume a non-rotating blade modelled as beam (1D continuum) where each point in the beam's domain has assigned a bending w(x,t) and twist θ(x,t) deformation. As before, consider the dynamic equilibrium of an arbitrary blade segment *dx* located within the beam domain, i.e., *eR<x<x+dx<R*:

Arbitrary segment FBD – perspective



Torque equilibrium about e.c. axis:

$$-Q + Q + dQ - m\,dx\,(\ddot{w} + e\ddot{\theta})e + L\,dx\,e_A + M_A dx - I_{c.m.}\ddot{\theta}\,dx = 0$$

The resulting twist equation due to combined bending and twist:

$$Q' - (I_{c.m.} + me^2)\ddot{\theta} - me\ddot{w} = -(Le_A + M_A)$$

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Coupled bending-twist problem

Use previously discussed constitutive relationships for bending and torsion:

$M = EI\,w''$ … Euler-Bernoulli bending theory

$Q = GJ\,\theta'$ … St. Venant's torsion theory

$I_{e.c.} = I_{c.m.} + me^2$ … sectional quadratic moment of area about e.c.

Resulting coupled bending-twist equations for nonrotating blade:

$$(EI\,w'')'' + m\ddot{w} + me\ddot{\theta} = L \quad \text{… PDE order = 4}$$

$$(GJ\,\theta')' - I_{e.c.}\ddot{\theta} - me\ddot{w} = -(Le_A + M_A) \quad \text{… PDE order = 2}$$

We can use matrix notation to highlight the coupled nature of the resulting bending and twist equations:

$$\underbrace{\begin{bmatrix} EI(\circ)'' & 0 \\ 0 & GJ(\circ)' \end{bmatrix}}_{\text{Elastically uncoupled}} \begin{bmatrix} w'' \\ \theta' \end{bmatrix} + \underbrace{\begin{bmatrix} m & me \\ -me & -I_{e.c.} \end{bmatrix}}_{\text{Inertially coupled}} \begin{bmatrix} \ddot{w} \\ \ddot{\theta} \end{bmatrix} = \underbrace{\begin{bmatrix} L \\ -(Le_A + M_A) \end{bmatrix}}_{\text{Aerodynamically coupled}}$$

This is a PDE system of order 6 which will require specification of ("balanced") 4+2=6 boundary conditions. These conditions, specified at the root and the tip of the blade (or wing) will involve geometric or dynamic (force) conditions.

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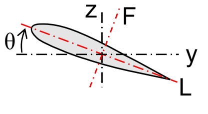
Examples of other sources of coupling effects

Blade pitch/pretwist



Coupling between flap and lead-lag due to non-zero pitch angle (pre-twist, twist, …):

$$M_z' + S_z = 0$$

$$M_y' + S_y = 0$$

where:

$$\begin{bmatrix} M_z \\ M_y \end{bmatrix} = \begin{bmatrix} A & C \\ C & B \end{bmatrix} \begin{bmatrix} w'' \\ v'' \end{bmatrix}$$

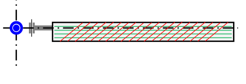
$$A = EI_F + (EI_L - EI_F)\sin^2\theta$$

$$B = EI_L + (EI_F - EI_L)\sin^2\theta$$

$$C = (EI_L - EI_F)\sin(2\theta)/2$$

The problem of unsymmetrical bending.

Composites



Coupling between flap bending and twist due to non-zero bending-twist coupling:

$$M' + S = 0$$

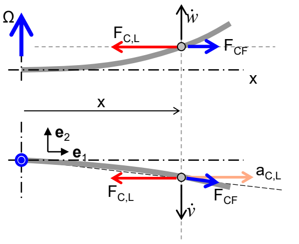
$$Q' - M_\theta = 0$$

where:

$$\begin{bmatrix} M \\ Q \end{bmatrix} = \begin{bmatrix} EI & -K \\ -K & GJ \end{bmatrix} \begin{bmatrix} w'' \\ \theta' \end{bmatrix}$$

This formulation represents the coupled bending-twist *constitutive law* where *K* represents strength of the bending-twist coupling. Such coupling can be introduced by a particular composite layup.

Coriolis



Coupling between flap and lead-lag due to non-zero Coriolis acceleration:

$$a_{C,L} = 2\Omega\dot{v}$$

$$F_{C,L} = -2\Omega\dot{v}\,m\,dx$$

$$F_{CF} + F_{C,L} = (\Omega^2 x - 2\Omega\dot{v})\,m\,dx$$

CF and Coriolis term for flapping:

$$-\left(w'\int_x^R \Omega^2 m\tilde{x}\,d\tilde{x}\right)' + \underbrace{\left(w'\int_x^R 2\Omega m\dot{v}\,d\tilde{x}\right)'}_{\text{Nonlinear term (inconvenient)}}$$

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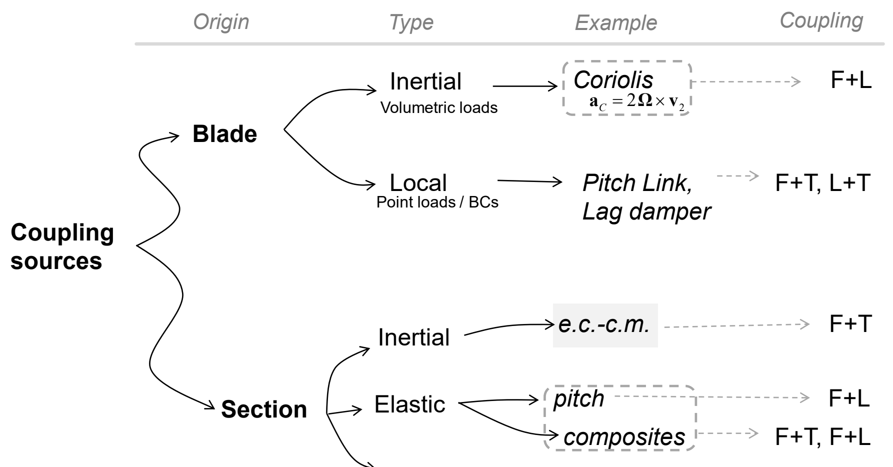
Sources of coupling in 1D structures

Previously introduced and mentioned bending-twist PDE models are typically considered in broader context of various other possible coupling sources:



The choice of a "correct" model depends on the context, applications or problem constraints and required model "fidelity".

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Summary

- Resulting bending-twist PDE model (EoMs) represents local point-wise and differential equilibrium between the internal reaction, inertial and applied (e.g., aerodynamic) loads.
- The formulated bending twist-model uses Euler-Bernoulli bending and St. Venant's torsion constitutive relationships to establish the link between the internal reaction loads and system's deformation state (2-DOFs per beam "section").
- The established bending-twist model is represented by a PDE system of the 6^{th} order. This system requires two bending-specific and one torsion-specific boundary condition for each side of the solved blade domain.

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